

Solutions – Quiz 1

ECON 300: Labour Economics

Instructor: Muhebullah Karimzada

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Multiple Choice (5 points)

1. **Answer: (c).** A worker on temporary layoff expecting recall is counted as unemployed (they are without work but still attached to the labour market). A discouraged worker who stopped searching is *not* counted as unemployed because they are not actively searching.
2. **Answer: (b).** The LFPR is:

$$LFPR = \frac{\text{Labour Force}}{\text{Working-age population}},$$

and the labour force equals employed plus unemployed.

3. **Answer: (c).** A higher wage raises the opportunity cost of leisure: one more hour of leisure means giving up more earnings.
4. **Answer: (b).** The substitution effect makes leisure relatively more expensive, so individuals substitute away from leisure toward work, increasing hours worked.
5. **Answer: (c).** A stronger preference for leisure increases the wage required to induce the person to give up leisure for work, raising the reservation wage.

Problem 1: Labour Force Statistics and Interpretation (7.5 points)

Given:

$$\text{Working-age population} = 1,000,000, \quad E = 640,000, \quad U = 60,000, \quad N = 300,000.$$

(a) Labour force and LFPR (2.5 points)

Step 1: Compute the labour force. By definition,

$$LF = E + U = 640,000 + 60,000 = 700,000.$$

Step 2: Compute the labour force participation rate.

$$LFPR = \frac{LF}{\text{Working-age population}} = \frac{700,000}{1,000,000} = 0.70 = 70\%.$$

(b) Unemployment rate and employment-to-population ratio (2.5 points)

Step 1: Unemployment rate.

$$u = \frac{U}{LF} = \frac{60,000}{700,000} = 0.085714 \dots \approx 8.57\%.$$

Step 2: Employment-to-population ratio.

$$\frac{E}{\text{Working-age population}} = \frac{640,000}{1,000,000} = 0.64 = 64\%.$$

(c) Recession with job losses and discouraged workers (2.5 points)

Shock 1: 20,000 employed lose jobs. Those workers move from employed to unemployed (initially).

$$E' = 640,000 - 20,000 = 620,000, \quad U' = 60,000 + 20,000 = 80,000.$$

Shock 2: 15,000 discouraged workers stop searching. This does not affect the labour force. They were not part of labour force and still they are not part of it.

$$U'' = 80,000$$

Since they exit the labour force, the new labour force becomes:

$$LF'' = E' + U'' = 620,000 + 80,000 = 700,000.$$

New unemployment rate:

$$u'' = \frac{U''}{LF''} = \frac{80,000}{700,000} = 0.11429 \dots \approx 11.43\%.$$

Problem 2: Labour Supply, Market Equilibrium, and Minimum Wage (7.5 points)

Given individual problem:

$$C = 600 + 25h, \quad U = C - 0.5h^2.$$

Given market:

$$L^D = 120 - 2w, \quad L^S = 20 + 3w.$$

(a) Optimal hours worked (2.5 points)

Step 1: Substitute the budget constraint into utility.

$$U(h) = (600 + 25h) - 0.5h^2 = 600 + 25h - 0.5h^2.$$

Step 2: Differentiate with respect to h and set equal to zero (FOC).

$$\frac{dU}{dh} = 25 - h.$$

Set FOC:

$$25 - h = 0 \quad \Rightarrow \quad h^* = 25.$$

Step 3: Check second derivative (SOC).

$$\frac{d^2U}{dh^2} = -1 < 0,$$

so $h^* = 25$ is a maximum.

Answer: The worker supplies **25 hours per week**.

(b) Market equilibrium wage and employment (2.5 points)

Step 1: Set supply equal to demand.

$$120 - 2w = 20 + 3w.$$

Step 2: Solve for w .

$$120 - 20 = 3w + 2w \quad \Rightarrow \quad 100 = 5w \quad \Rightarrow \quad w^* = 20.$$

Step 3: Substitute back to get equilibrium employment. Using demand:

$$L^* = 120 - 2(20) = 120 - 40 = 80.$$

(Check with supply: $20 + 3(20) = 20 + 60 = 80$.)

Answer: Equilibrium wage is $w^* = 20$ and employment is $L^* = 80$.

(c) Minimum wage $w = 20$ (2.5 points)

Step 1: Compare minimum wage to equilibrium wage.

$$w_{\min} = 20, \quad w^* = 20.$$

Because the minimum wage equals the competitive equilibrium wage, it is **not binding**.

Step 2: Compute labour demanded and supplied at $w = 20$.

$$L^D(20) = 120 - 2(20) = 80, \quad L^S(20) = 20 + 3(20) = 80.$$

Step 3: Unemployment (excess supply).

$$U = L^S - L^D = 80 - 80 = 0.$$

Answer: The minimum wage is **not binding** and creates **zero unemployment**.